INCH-POUND

MIL-L-43498D
12 September 1990
SUPERSEDING
MIL-L-43498C
19 September 1983

MILITARY SPECIFICATION

LINER, COLD WEATHER TROUSERS, FIELD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a trouser liner made from quilted polyester batting material, for the cold weather field trousers.

1.2 Classification. The liner shall be of one type in the following sizes (see 6.2):

Schedule of sizes

Short and Regular	Long
X-Small	X-Small
Small	Small
Medium	Medium
Large	Large
X-Large	X-Large

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Untreated
A-A-50198	-	Thread, Gimp, Cotton, Buttonhole
A-A-50199	-	Thread, Polyester Core, Cotton- or Polyester Covered
V-B-871	-	Button, Sewing Hole, and Button, Staple, (Plastic)
JJ-W-155	-	Webbing, Textile, (Cotton, Elastic)
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-C-508	-	Cloth, Oxford, Nylon, 3 Ounce
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-B-41826	-	Batting, Synthetic Fibers, Polyester, (Unquilted and Quilted)

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

THE COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Card of America

Department of Defense Standard Color Card for Sewing Threads

(Color cards may be available from the Color Association of the United States, 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color smaples may be obtained from the contracting activity or as directed by the contracting activity.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Guide sample. Guide samples, when furnished, are solely for the guidance and information to the contractor (see 6.3). Variation from the document may appear in the sample, in which case this specification shall govern.

3.3 Materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Basic material. The basic material shall be quilted polyester batting, conforming to type IV, class 1, style a, of MIL-B-41826.

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3.3.2 Waistband and binding. The material for waist lining, side opening facings, and bias binding shall be nylon oxford cloth, dyed Olive Green 106, conforming to type I, class 2 of MIL-C-508, except that the colorfastness requirements to light and wet-drycleaning shall not apply.

3.3.3 Labels. Each liner shall have an identification label, a size label, and an instruction label conforming to DDD-L-20, except label color shall be Medium Green, Cable Nos. 70034, 70130, or 70131 (see 2.2). Fastness to laundering requirements shall apply.

3.3.3.1 Identification label. The identification label shall conform to type VI, class 1.

3.3.3.2 Size label. The size label shall conform to type VI, class 2 and shall contain the following information for each respective size:

X-Small - Short and Regular

Waist: Below 27 inches
Inseam: Up to 32-1/2 inches
Stock No:
NATO Size: 6783/5869

X-Small - Long

Waist: Below 27 inches
Inseam: Over 32-1/2 inches
Stock No:
NATO Size: 8390/5869

Small - Short and Regular

Waist: From 27 to 31 inches
Inseam: Up to 32-1/2 inches
Stock No:
NATO Size: 6783/6979

Small - Long

Waist: From 27 to 31 inches
Inseam: Over 32-1/2 inches
Stock No:
NATO Size: 8390/6979

Medium - Short and Regular

Waist: From 31 to 35 inches
Inseam: Up to 32-1/2 inches
Stock No:
NATO Size: 6783/7989

Medium - Long

Waist: From 31 to 35 inches
Inseam: Over 32-1/2 inches
Stock No:
NATO Size: 8390/7989

Large - Short and Regular

Waist: From 35 to 39 inches
Inseam: Up to 32-1/2 inches
Stock No:
NATO Size: 6783/8999

Large - Long

Waist: From 35 to 39 inches
Inseam: Over 32-1/2 inches
Stock No:
NATO Size: 8390/8999

X-Large - Short and Regular

Waist: From 39 to 43 inches
Inseam: Up to 32-1/2 inches
Stock No:
NATO Size: 6783/9909

X-Large - Long

Waist: From 39 to 43 inches
Inseam: Over 32-1/2 inches
Stock No:
NATO Size: 8390/9909

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3.3.3.3 Instruction label. The instruction label shall conform to type VI, class 3 and shall contain the following information:

LINER, COLD WEATHER TROUSERS, FIELD

1. Button into trousers for extra warmth.
2. Buttoning side openings for easy removal.
3. Dry liner and trousers separately.
4. Avoid contact with heated stoves, open fires or other heat sources; fabric may fuse.
5. Wash by hand or machine in warm water with mild soap or detergent, rinse thoroughly, and drip dry or machine dry.

DO NOT DRY CLEAN.

DO NOT REMOVE THIS LABEL

3.3.3.4 Combination size and identification label. The identification label (class 1) and the size label (class 2) may be combined into one label conforming to type VI, classes 1 and 2 combined.

3.3.3.5 Combination size, identification and instruction label. The identification label (class 1), size label (class 2) and instruction label (class 3) may be combined in one label conforming to class 14. The coating requirement shall conform to class 3 label.

3.3.4 Thread. The thread shall be cotton or polyester covered polyester core conforming to A-A-50199, ticket number 30, 2 or 3 ply or 50-2 ply for stitching and seaming, 70-2 for overedging, and 30, 2 or 3 ply for buttonholes and button sewing. The thread shall be dyed Olive Drab S-1 C.A. 66022.

3.3.4.1 Colorfastness. The dyed thread shall show fastness to laundering equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to laundering.

3.3.5 Gimp. The cotton gimp for reinforcing buttonholes shall be no. 8, dyed Olive Drab S-1 C.A. 66022, conforming to A-A-50198.

3.3.5.1 Colorfastness. The dyed gimp shall show fastness to laundering equal to or better than the standard sample. When no standard sample is available, the dyed gimp shall show "good" fastness to laundering.

3.3.6 Buttons. The buttons shall be type II, class D, style 26, 30-line of V-B-871. The buttons shall be Olive Green BP, C.A. 62016 with a dull finish.

3.3.7 Elastic. The elastic for the waistband shall be type I bleached or unbleached, class 6 of JJ-W-155.

3.4 Design. The design of the liner shall be a button in liner with buttoned side openings for easy removal (see figure 1).

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3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government. The working patterns shall be identical to the standard patterns which shall not be altered in any way. The patterns provide 1/2 inch seam allowance for all seams (to finish 3/8 inch wide).

3.5.1 Pattern parts. The component parts shall be cut from the materials specified below in accordance with pattern parts indicated in table I.

TABLE I. List of pattern parts

Material	Pattern nomenclature	Cut parts
Basic material	Front	2
	Back	2

3.6 Construction. The construction shall conform in all respects to the requirements specified in table II and herein. Figure 1 is furnished solely for guidance and information. Should variation from this specification appear in figure 1, this specification shall govern.

3.6.1 Stitches, seams, and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table II. Seam allowance shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers, or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the looper (underthread) may be on the outside or the inside of the liner. The reverse side of the seam type LSbm-3 (overedge side) shall be on the outside of the liner. The side of the liner worn nearest the body shall be considered the inside.

3.6.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/4 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewed.

3.6.1.1.1 Repairs of type 301 stitching.

3.6.1.1.1.1 Quilty batting stitching. Broken stitches of the quilting pattern may be repaired by stitching back of each end of break not less than 1 inch.

3.6.1.1.1.2 Liner fabrication stitching.

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a. When thread breaks or bobbin run-outs occur during stitching, except prestitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Except for prestitching, thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.6.1.2 Type 401, 503, 504, 515, 516, 517, 518 and 519 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.6.1.1.1.2a and 3.6.1.1.1.2b. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering or the materials sewn.

3.6.2 Buttonholes. The buttonholes shall be the eyelet-end-taper-bar-type worked over gimp (see 3.3.5) with not less than four tacking stitches at bar end (not counting cross-over stitch). The finished cut length shall be 3/4 to 7/8 inches. The buttonholes shall be clean cut with the stitching securely caught in the fabric.

3.7 Manufacturing operations requirements. The liner shall be manufactured in accordance with operations specified in table II. The contractor is not required to follow the exact sequence of operations. Any additional basting or holding stitching may be used providing the threads are removed or are not visible on the finished liner.

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NO.	TABLE II. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut the liner in strict accordance with patterns furnished, which show size, directional lines, and notches for proper assembly of all parts. The directional lines shall be in the warp direction. b. Cut the nylon oxford for waistband lining and side opening facings in the warp direction and of sufficient length and width to conform to the requirements specified in operations 10 and 11. c. Cut the nylon oxford for the binding on the bias in sufficient width to conform to the requirements specified in operation 7. d. Pre-cut elastic for waistband as follows: Size — Length — Cut part X-small — Front 3-1/8" — 2 Back 10-1/2" — 1 Small — Front 4-1/8" — 2 Back 12-1/2" — 1 Medium — Front 5-1/8" — 2 Back 14-1/2" — 1 Large — Front 6-1/8" — 2 Back 16-1/2" — 1 X-Large — Front 7-1/8" — 2 Back 18-1/2" — 1						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd) NOTE: Care shall be exercised when cutting the basic material and handling the cut parts as the quilting stitches have a tendency to run back.						
2.	Replacement of damaged parts. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2 shall be removed from production and replaced with non-defective and properly matched components.						
3.	Component marking. Mark all components of the liner to insure a uniform size throughout the liner. Any method of marking may be used except: a. Metal fastening devices. b. Sew on type tickets. c. Adhesive type tickets which leave traces of adhesive on the material upon removal of the ticket.						
4.	Labels.						
	a. Sew instruction, size and identification labels (on all four sides not more than 3/16 inch from edges) to inside center (1 inch off center tolerance) of right back as follows:	301	LSbj-1	10-14	50	50	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Labels. (cont'd) (1) Size label - top edge 2-1/2 to 3 inches below top edge of waist. (2) Identification label - may abut or be caught in bottom line of stitching of the identification label. (3) Instruction label - may abut or be caught in bottom line of stitching of the identification label. (4) Combination label (when applicable) - shall be positioned as in (1) above.						
5.	Join crotch seam, seat seam, and inseams. Finished appearance. The overedge stitched edges shall finish on the outside of the liner. The bright of overedge stitching shall be 3/16 to 1/4 inch gage.						
	a. Join crotch seams and seat seams by overedge stitching and seaming the raw edge together in one operation, respectively.	515 or 516 or 519 or 517 or 518	LSmb-3(a) LSbm-3(a)	10-14 10-14	50 (chain stitching) 70 (overedge stitching) 50 (lock stitching) 70 (overedge stitching)	70 70 50 70	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Join crotch seam, seat seam, and inseams. (cont'd)						
	b. Fold pleats on leg inseams as indicated by marks on pattern and join inseams by overedge stitching and seaming the raw edges together in one continuous operation. The fold of pleats shall be towards the bottom on inside of liner	515 or 516 or 519 or	LSmb-3(a)	10-14	50 (chain stitching) 70 (overedge stitching)	50 (chain stitching) 70 (overedge stitching)	
		517 or 518	LSmb-3(a)	10-14	50 (chain stitching) 70 (overedge stitching)	50 (chain stitching) 70 (overedge stitching)	
	c. For stitch types 515, 516, or 519. Turn seam allowances of joining seams to one side and raise stitch 3/16 to 1/4 inch from folded edges.	301 or 401	LSmb-3(b)	10-14	50	50	
6.	Overedge stitch raw edges.						
	Overedge stitch raw edges around leg bottoms, waist, fly opening and side openings.	503	EFd-1	8-10	70	70	
7.	Bind fly opening and bottom of legs.						
	Finished appearance. The fly opening and leg bottom binding shall finish 3/8 to 5/8 inch wide with no raw edges.						
	a. Bind the overedge stitched edges of fly opening and leg bottoms with the raw edges of the binding turned in and caught in the stitching, and the raw ends starting and finishing at top edges of waist and outside edges of leg bottoms.	301	BSc-1	10-14	50	50	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Attach back buttonholes reinforcements, (as indicated on patterns).						
	a. Place a small square of nylon oxford cloth (large enough to carry the buttonhole yet small enough so the excess will not roll) on the inside of back liner marked for two center button holes and stitch on all four sides.	301		10-14	50	50	
	b. Make two center buttonholes as marked, through reinforcements.	Button-hole		52-56 per button-hole including tack	30	30	
9.	Attach waistband elastic (front and back).						
	a. Turn ends of elastic 3/8 inch, position as marked on patterns, and stitch across twice, 1/8 inch from edge.	301	LSd-2	10-14	50	50	
10.	Attach waistband linings (front and back).						
	Finished appearance. The waistband lining shall finish 2 to 2-1/4 inches in width on the inside and 1/2 to 5/8 inch in width on the outside of liner. Ends of back waistband lining shall finish even with side openings. Ends of front waistband linings shall finish even with side openings and extend both sides of fly opening 2-3/4 to 3 inches in length to be turned to the outside.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Attach waistband linings (fronts and backs). (cont'd)						
	a. Turn in bottom edge of lining and stitch to the inside of liner 1/16 to 1/8 inch from folded edge. The line of stitching shall continue the length of the lining.	301 or 401	LSd-1 and EFa-1	10-14	50 50	50 70	
	b. Turn top edge of lining to the outside of liner and, with the raw edge turned in, single stitch 1/16 to 1/8 inch from folded edge. The line of stitching shall continue the length of the lining. or	301 or 401	SSl-1 and EFa-1	10-14	50 50	50 70	
	c. Seam bottom edge of lining to the outside of liner; turn and raise stitch 1/16 to 1/8 inch from turned edge, with the stitching continued the length of the lining.	301 or 401	LSbk-2 and EFa-1	10-14	50 50	50 70	
	d. Turn top edge of lining to the inside of liner and with the raw edge turned in, single stitch 1/16 to 1/8 inch from turned edge. The line of stitching shall continue the length of the lining. or	301 or 401	LSd-1 and EFa-1	10-14	50 50	50 70	
	e. Attach lining to liner in one operation using a double needle machine 1-1/2 to 1-3/4 inch gage, with the lines of stitching continuing the length of the lining.	301 or 401	SSl-1 LSd-1 and EFa-1	10-14	50 50	50 70	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Attach waistband linings (fronts and backs). (cont'd)						
	f. Fold extending ends of waist lining to the outside of liner and with the raw ends turned in, single stitch 1/16 to 1/8 inch from all edges.	301	LSbj-1 and SSa-1 and OSf-1	10-14	50	50	
		or 401	and LSd-1		50	70	
11.	Attach side opening facings (front and back).						
	Finished appearance. The facings for the front side openings shall finish 2-1/8 ± 1/8 inches in width to both the inside and outside of liner. Back side opening facings shall finish 1 ± 1/8 inch in width to both the inside and outside of liner. All raw edges shall be turned to the inside of facings.						
	a. Stitch right side of facing to inside of liner with edge of liner at half width point of facing and continue the length of liner including pleats as indicated on patterns.	301 or 401	LSd-1	10-14	50	50	
			EFa-1		50	70	
	b. Turn facing to the outside of liner and turning raw edges under stitch 1/16 to 1/8 inch from folded edge across ends and length of facing in one continuous stitching tacking both ends 1/2 inch.	301 or 401	SS1-1	10-14	50	50	
			EFb-1		50	70	
12.	Make buttonholes.						
	a. Make four more vertical buttonholes in waistband, as indicated by marks on pattern, with the inside edges of eyelets 1/2 to 3/4 inch from top edge of waist. The purling shall finish on the inside of liner.	Buttonhole		52-56 per buttonhole including tack	50	50	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Make buttonholes. (cont'd)						
	b. Make three horizontal buttonholes on fronts as indicated on patterns (one on left, two on right front). Purling shall be on the inside of the liner.	Button-hole		52-56 per button-hole including tack	30	30	
	c. Make a horizontal buttonhole located 1 inch up from the leg bottom. Locate and make four horizontal buttonholes equally spaced between the waist buttonhole and the leg bottom buttonhole. The edge of the buttonhole eyelet shall be 1/2 inch from the edge of the facing. Purling shall be on the inside of the liner.	Button-hole		52-56 per button-hole including	30	30	
13.	Attach buttons.						
	a. Sew one 30-line button to left front fly opening positioned 1 to 1-1/4 inches from finished front edge and centered (1/4 inch tolerance) on waistband. Measurements to be taken from center of button.	301 or		14-16 per button	30	30	
		101		14-16 per button	30		
	b. Sew 30-line buttons to outside of back side facings matching them to buttonholes on front side facings and 3/4 inch from finished edge, catiching lining all plies.	301 or		14-16 per button	30	30	
		101			30		

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					NEEDLE	BOBBIN/ LOOPER	COVER
14.	Clean liner. a. Trim and remove all thread ends, except ends of quilting stitching on the basic material. b. Remove all spots, stains and size markings.						

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3.8 Finished measurements. The finished liner shall conform to measurements for each respective size, as shown in table III. Measurements shall be taken after the liner has been smoothed flat (applying slight tension) along the dimension to be measured. During the measurement, however, no tension whatsoever shall be applied to the end item.

TABLE III. Finished measurements

Size	1/2 Waist 1/	Inseam (short and regular) 2/	Inseam (long) 2/
	Inches	Inches	Inches
X-Small	13-1/2	22	26
Small	15-1/2	22	26
Medium	17-1/2	22	26
Large	19-1/2	22	26
X-Large	21-1/2	22	26
Tolerance	+ 3/4 - 1/2	± 3/4	± 3/4

1/ Measurement taken with the waist buttoned, and the liner laid flat and smooth, with elastic relaxed. Measure across front at the center of the waistband from side edge to side edge.

2/ Measure along inside seam of liner from end of crotch to bottom of leg.

3.9 Workmanship. The finished liner shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspection are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or

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supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item visual examination. The end items shall be examined for the defects listed in table IV. The lot size shall be expressed in units of liners. The sample unit shall be one liner. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects, 15.0 for major and minor A combined defects, and 40.0 for total (major, minor A, and minor B combined) defects.

TABLE IV. End item visual defects

Defect	Classification Major	Minor A	Minor B
Material defects and damages			
a. Any hole, cut, tear, needle chew, smash, or multiple float in any ply	101		
b. Darn or mend in any ply		201	

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TABLE IV. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Material defects and damages (cont'd)			
c. Exposed drill hole in any ply			301
d. One or more rows of stitching omitted in the quilting	102		
e. More than 1-1/2 inches of broken or missing stitches in the quilting			302
f. Thick or thin places	103		
Cleanness			
a. Any spot or stain		202	
b. Thread ends not trimmed or loose threads not removed on major portion of liner, except thread ends of quilting on basic material			303
c. One or more size markings not removed			304
Components and assembly			
a. Any component part or required operation omitted (unless otherwise classified herein)	104		
b. Any operation not performed as specified (unless otherwise classified herein)			305
Cutting			
Any component part not cut in conformance with directional lines indicated on pattern or not cut in accordance with specification requirements	105		
Seams and stitching			
a. Accuracy of seaming:			
1. Seams twisted, puckered, or pleated		203	
NOTE: Nylon fabric has a tendency to pucker at joining seams and stitchings. The puckering when not so excessive as to seriously affect appearance shall not be classified as a defect			
2. Part of liner caught in any unrelated operation or stitching	106		

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TABLE IV. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
3. Ends of seams or stitching produced with 301 stitch type, when not caught in other seams or stitching, backtacked less than 1/4 inch		204	
4. Stitching overlapping ends of thread break less than 1/2 inch, or broken ends of quilting stitching stitched back of breaks less than 1 inch			306
5. Ends of a continuous line of stitching not overlapped or overlapped less than 1/2 inch			307
b. Gage of stitching (raised, edge or double stitching):			
1. Beyond range of width specified or varies more than 1/16 inch when no range is specified			308
2. Irregular, i.e., unevenly gaged or various stitchings not uniform (to be scored only when the condition exists on major portion of seam)			309
c. Open seams: (not applicable to quilting stitching)			
- up to 1/4 inch inclusive			310
- more than 1/4 inch to 1 inch inclusive		205	
- more than 1 inch	107		
NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more continuous skipped or runoff stitches occur. On double stitched seams, a seam is classified as open when either one side or both sides of seams are open			
d. Raw edges (excluding overedge stitching and end binding, when not turned under):			
- up to 1/4 inch inclusive			311
- more than 1/4 inch to 1 inch inclusive		206	
- more than 1 inch	108		
Any raw edge caused by omission of overedge stitching for a distance of 3 inches or more		207	

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TABLE IV. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
NOTE: Raw edges are classified as such when they occur at places where edge is required to be turned under or overedge stitched. Raw edges not securely caught in stitching shall be classified as open seams			
e. Runoffs:			
1. Joining seams - when resulting in an open seam, use "Open seam" classification			
2. Edge or raised stitching (when not resulting in an open seam):			
- 1/4 inch up to 1/2 inch inclusive			312
- more than 1/2 inch		208	
f. Seam and stitch type:			
1. Not specified seam or stitch type		209	
2. Overedge stitched edge on joining seams not finished on the outside of liner		210	
g. Stitch tension:			
1. Loose tension resulting in a loose seam	109		
2. Loose tension on raised or edge stitching		211	
3. Tight tension, i.e., stitches break when normal strain is applied to seam or stitching	110		
NOTE: Puckering is evidence of tight tension. When puckering is evident, the seam shall be tested by exerting pull in the lengthwise direction of seam			
h. Stitches per inch (to be scored only when the condition exists on the major portion of seam) (except overedge stitching):			
Less than minimum specified:			
- one or two stitches			313
- three stitches		212	
- four or more stitches	111		

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TABLE IV. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
One or more stitches less than minimum specified on overedge stitching			314
One or more stitches more than specified			315
1. Stitches skipped or broken on edge or raised stitching			
- 1/4 inch up to 1/2 inch inclusive			316
- more than 1/2 inch		213	
Buttonholes (construction)			
a. Stitching not caught in fabric causing stitching to pull away from fabric	112		
b. Wrong type, incomplete stitching, or uncut buttonhole		214	
c. One or more broken stitches in:			
- one buttonhole		215	
- two or more buttonholes	113		
d. Gimp omitted in one or more buttonholes			317
e. Cut length less than 3/4 inch or more than 7/8 inch			318
f. End stitched with less than four stitches per bar end (see 3.6.2):			
- one buttonhole		216	
- two or more buttonholes	114		
Front or side opening			
Opening uneven in length by:			
- 1/4 inch up to 3/8 inch inclusive			319
- more than 3/8 inch		217	
Binding			
a. Binding not caught to fabric, i.e., can be easily pulled away from edge of opening	115		
b. Binding shor, causing excessive puckers or pleats		218	
c. Binding less than 3/8 inch or more than 5/8 inch in width			320

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TABLE IV. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Waistband lining			
a. Construction:			
1. Twisted	219		
2. Width on inside of liner less than 2 inches or more than 2-1/4 inches			321
3. Width on outside of liner less than 1/2 inch or more than 5/8 inch			322
4. Irregular in width by more than 1/4 inch			323
5. Extended ends not turned outside of liner	220		
Side facings			
a. Width of facing not as specified			324
b. Irregular in width by more than 1/4 inch			325
c. Raw edges or edges not caught in stitching	221		
Pleats			
Fold of pleats not toward bottom on inside of liner			326
Inseams			
a. Ends of inseams staggered more than 3/8 inch (center to center)			327
Bottons			
a. Missing, loose, or broken	222		
b. Wrong size, type, or color	223		
c. Waistband buttons less than 1 inch or more than 1-1/4 inches from finished edge of left front; or off center on waist lining by more than 1/4 inch (measured from center of button)			328
d. Button stitching not locked at end of stitching cycle (tug at the loose end of thread, when accessible, to determine if it will ravel)	224		
e. Buttons and buttonholes out of alignment causing noticeable bulge when liner is buttoned	225		

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TABLE IV. End item visual defects (cont'd)

Defect	Classification Major	Minor A	Minor B
Position of horizontal buttonhole			
a. Inside edge of eyelet not required measurement from finished edge of right front and front side openings			329
b. Off center on waist lining by more than 1/4 inch			330
c. Purling not finishing on the outside of liner			331
Vertical buttonholes at waist			
a. Inside edge of eyelets less than 1/2 inch or more than 3/4 inch from top edge of waist			332
b. Purling not finishing toward the body			333
Labels (size, identification, instruction or combination, when applicable):			
a. Identification label missing, incorrect or illegible		226	
b. Size, instruction or combination (when applicable) label missing, incorrect, or illegible	116		
c. Off center on right back by more than 1 inch (as worn)			334
d. Not stitched on all sides or not positioned as specified			335

4.4.3 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified in table III. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of liners. The sample unit shall be one liner. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified

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Examine	Defect (cont'd)
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Each buttoned liner shall be laid flat and folded in half along the length with the front opening inside the fold. The liner shall be folded from the leg bottom up toward the waistband to an overall length of approximately 23 inches. Five folded liners of one size only shall be neatly stacked in a bundle, alternating waistbands end for end, and securely tied with cotton tape or twine approximately 4 inches in from each end.

5.1.2 Commercial preservation. Liners shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

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5.2.1 Level A packing. Fifteen liners of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A bundles shall be packed flat three in depth within the shipping container. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight draft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Fifteen liners of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A bundles shall be packed flat three in depth within the shipping container. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight draft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the box specification, except that the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. Liners preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), tunics packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 3 in accordance with appendix of MIL-STD-147.

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5.4 Marking. In addition to any special marking required by the contract or purchase order, units packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable. Each unit pack shall have attached a manila colored paper shipping tag for the identification marking.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this document.
b. Size(s) required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3 and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Subject term (key word) listing.

Cold/dry uniform
Cold protection
Inner garment
Warmth

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

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Navy - NU

Review activities:

Army - MD
Navy - MC
DLA - CT

Preparing activity:

Army - GL

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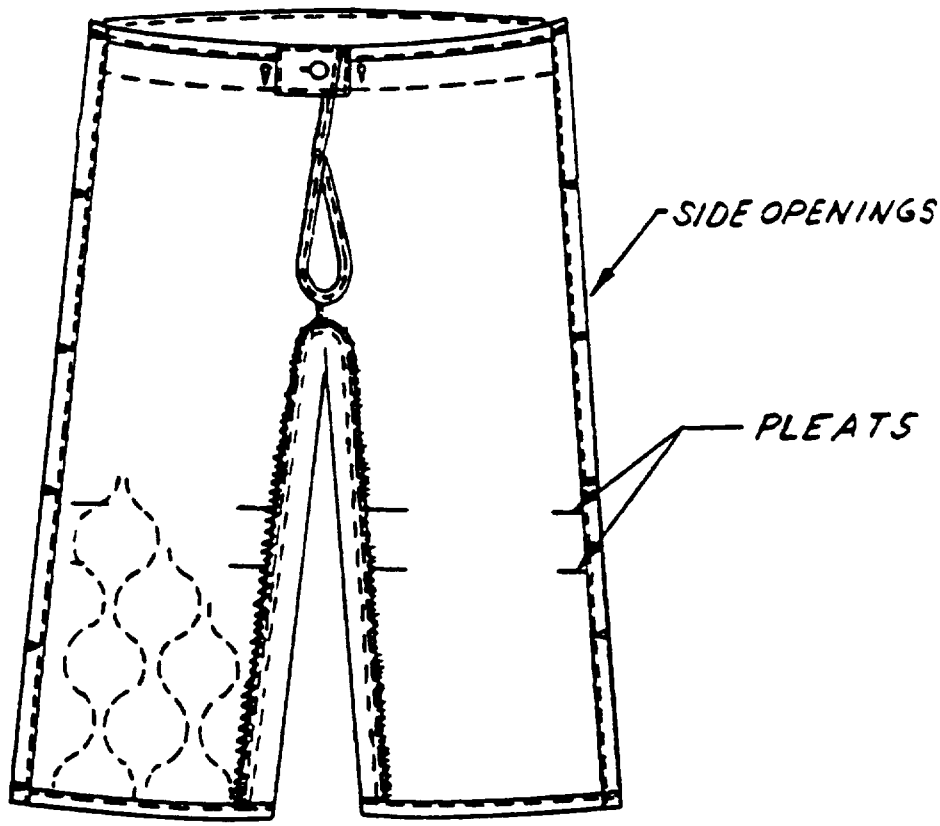


FIGURE 1 LINER, COLD WEATHER TROUSERS, FIELD

2-1-733

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-L-43498D	2 DOCUMENT DATE (YYMMDD) 1990 September 12

3 DOCUMENT TITLE
LINER, COLD WEATHER TROUSERS, FIELD

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)*

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U S Army Natick RD&E Center	508-651-5221	256-5221
c ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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